

Loan Approval Prediction Model

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Introduction



Business Problem: Build a model to predict loan approvals, helping banks reduce defaults and improve decision-making.



Why Important: Minimizes risks, speeds up loan processing, and enhances customer satisfaction.



Solution Idea: Use machine learning to analyze borrower data and predict outcomes reliably.



Our Approach: Conduct EDA, engineer features, test models, and optimize for accuracy.

Objective



Banks require an accurate way to predict loan approvals to minimize risks and improve lending efficiency.



We will identify key features influencing loan decisions.



Develop a machine learning model for accurate loan approval predictions.



Evaluate and optimize models to ensure practical and reliable outcomes.

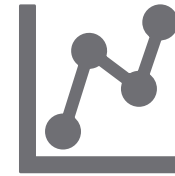
Dataset



Dataset was sourced from
Kaggle (Loan Approval
Classification Data)



Consists of 45,000 applicants
and their associated
information



The only errors we detected
within the dataset were
seven applicants with ages
over 100 and as high as 140.
These data points were
removed

Initial Conjectures



Younger applicants seek loans for education, older applicants seek loans for medial reasons



Applicants that own their home rather than mortgage or renters have a lower credit score because of less regular payment opportunities



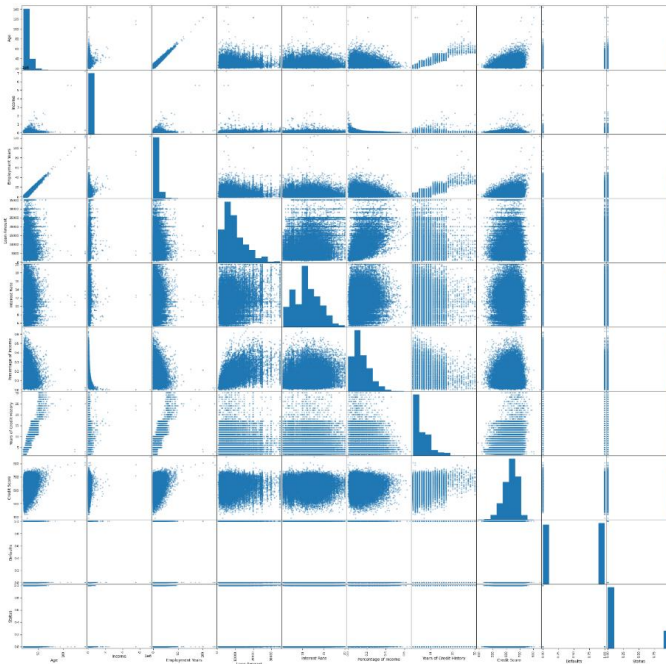
We can accurately predict a customer's credit score based on demographic and past financial information



We can create a model that can be used to predict whether an applicant's loan request will be accepted or rejected

Data Exploration

- **Features:** Age, Sex, Education, Income, Years Employed, Home Ownership, Loan Amount, Loan Intent, Interest Rate, Percentage of Income, Years of Credit History, Credit Score, Defaults, Status

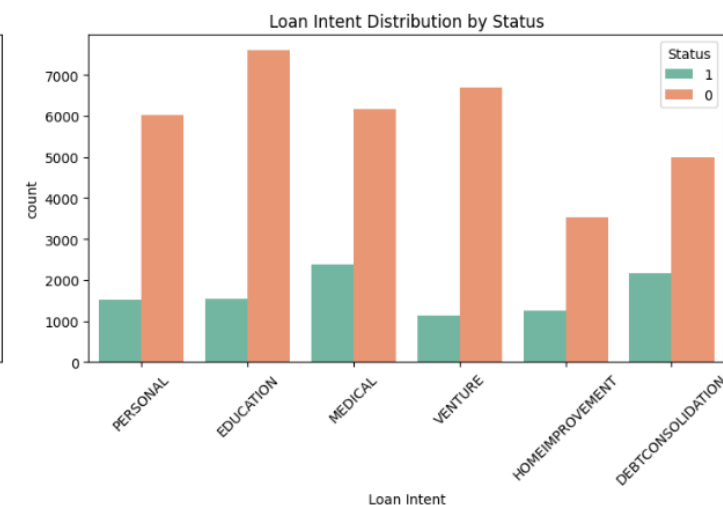
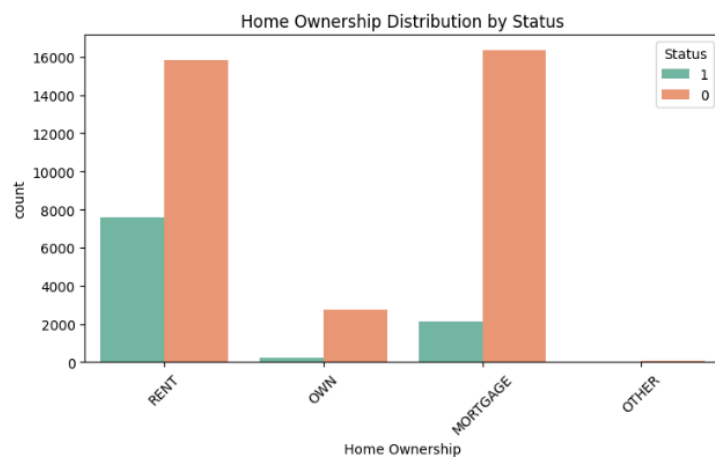
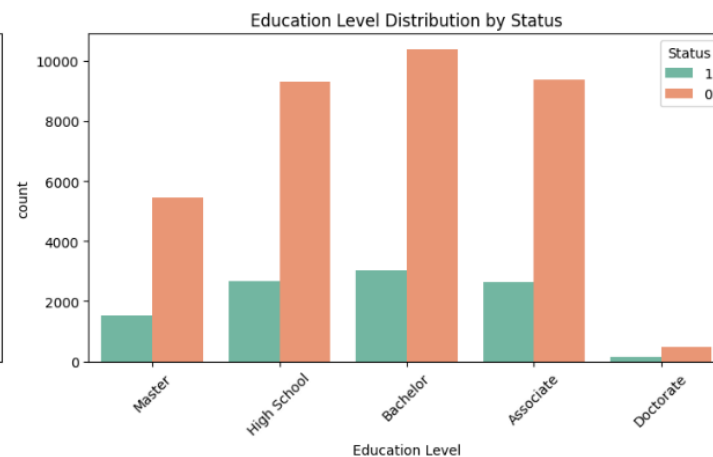
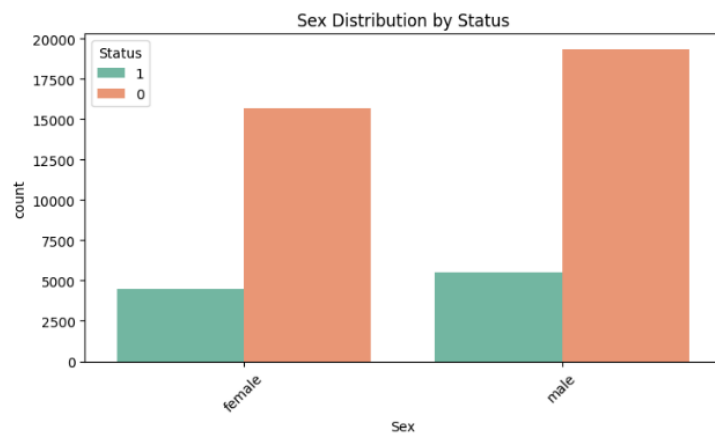


Numeric Variable Scatter Matrix



Numeric Variable Correlation Heatmap

Bar Charts of Categorical Variables by Status



Data Exploration Cont.

- About 22% of applicants were accepted
- Statistically significant relationship with acceptance for both home ownership and loan intent

Data Preprocessing

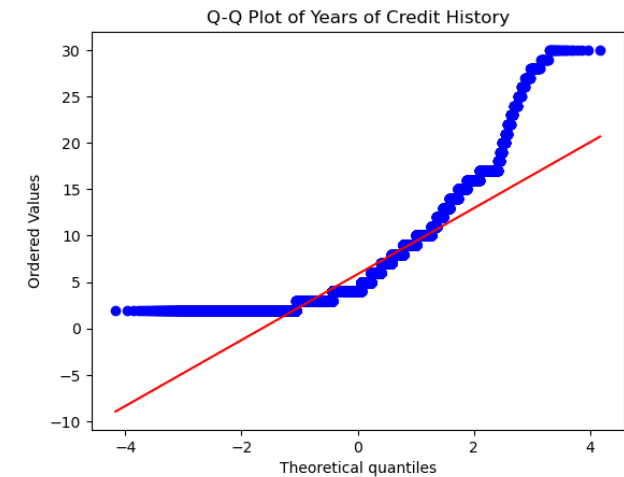
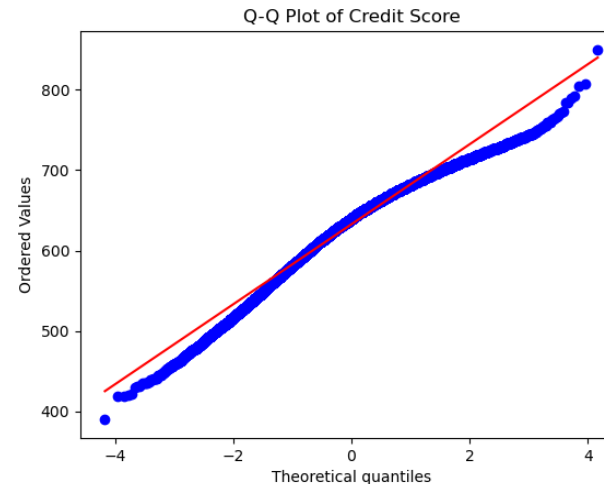


Split data into training, validation, and test data

Training set size: 28795
Validation set size: 7199
Testing set size: 8999



Sort numeric data into Gaussian and Pareto distributions and adjust accordingly with scalers



Model Development

Features Selected

- Previous loan defaults
- Income
- Loan interest rate
- Loan's percentage of income
- Home ownership: renting
- Loan intent: venture

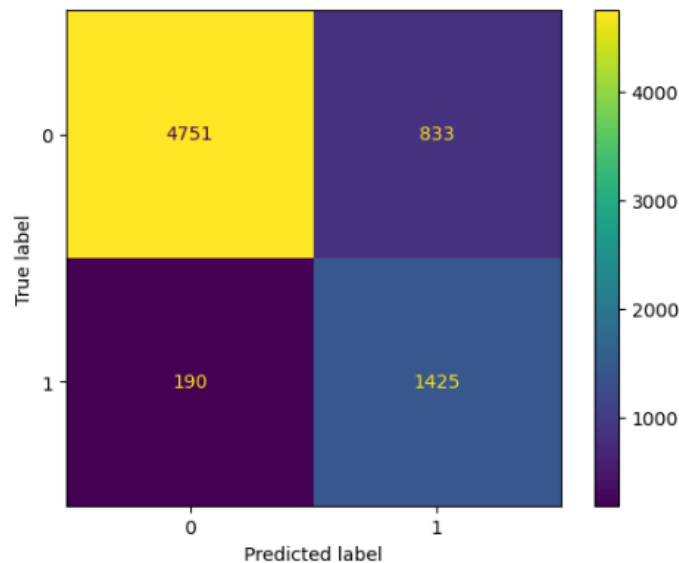
Models Chosen

- Logistic Regression
- Support Vector Classification
- Random Forest

Testing Models

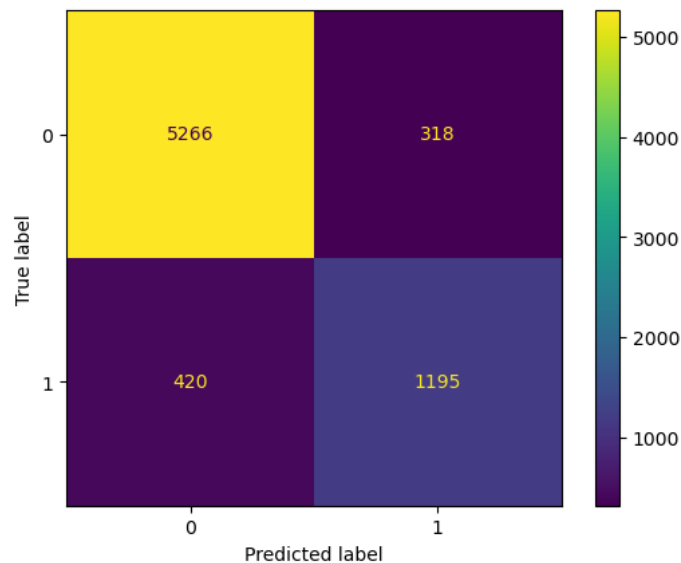
Logistic Regression

Accuracy Score
87.8%



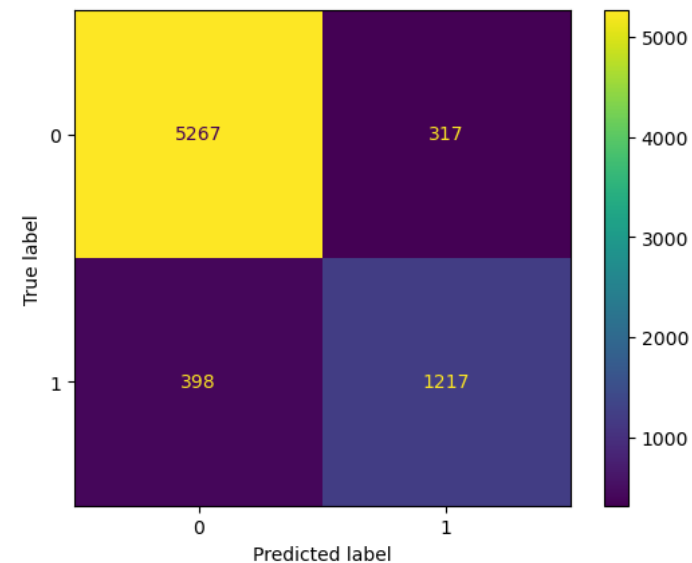
Support Vector

Accuracy Score
89.8%

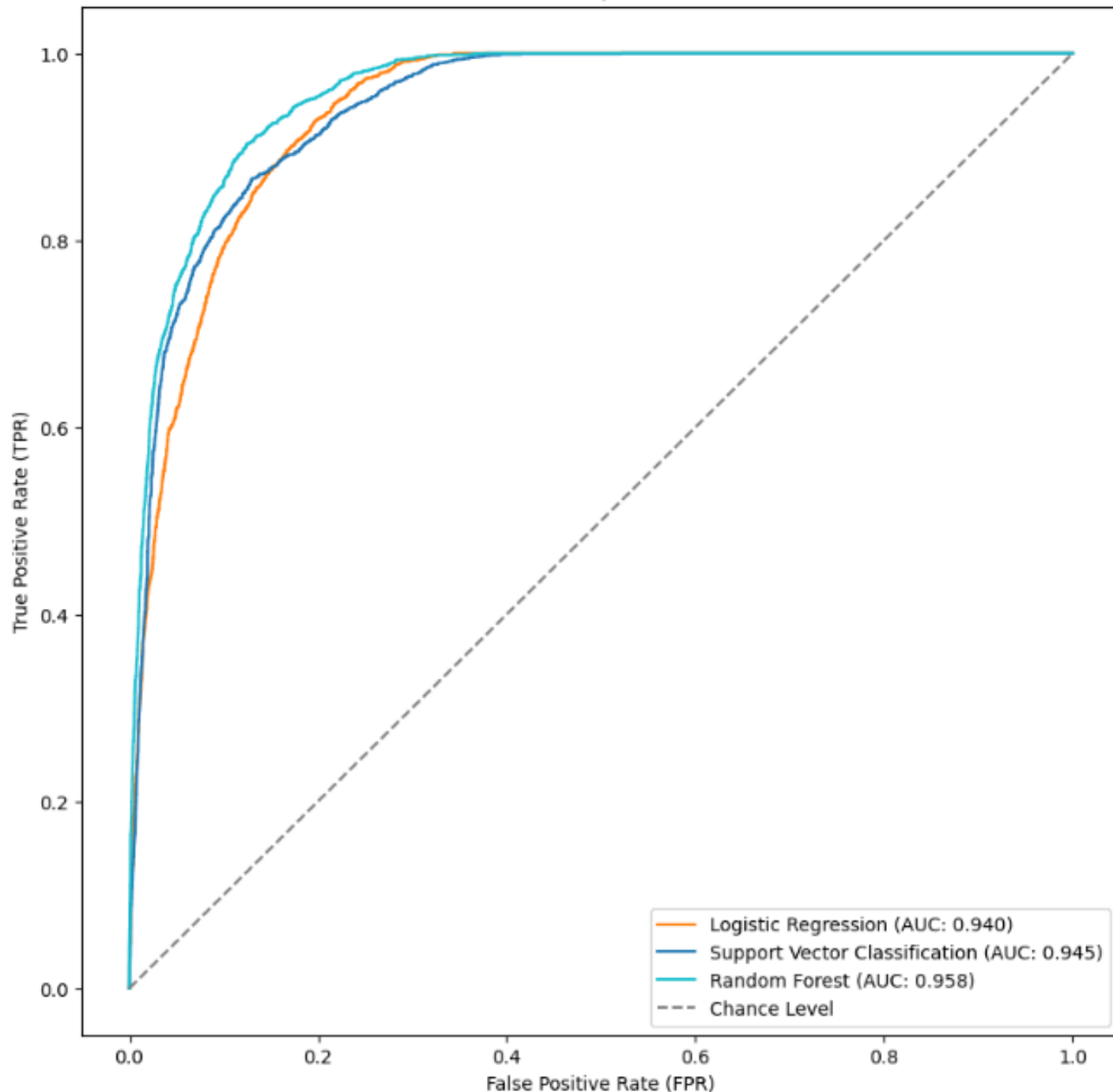


Random Forest

Accuracy Score
90.3%



ROC Curve with Optimal Thresholds



Model Selection

The Random Forest Model had the highest AUC value of 0.958

While the Support Vector Classification (0.945) and Logistic Regression (0.940) Models had lower AUC values

Due to Random Forest's Accuracy Score and ROC curve, this was chosen as our best model

Results & Conclusion

Conjecture 1

- Younger applicants prefer education loans; older applicants lean toward medical loans.
 - **Proved:** We found that younger applicants were more likely to pursue educational loans while older applicants pursued loans for medical reasons (as well as personal).

Conjecture 2

- Homeowners have lower credit scores due to fewer credit-building opportunities.
 - **Disproved:** The average credit score across home ownership categories were roughly equal.

Conjecture 3

- Income, employment, and education predict credit scores.
 - **Disproved:** We were unable to build a model that accurately predicted credit score based on the information we had. We suspect other specific information is needed.

Business Utilization



Efficiency of our model and its accuracy will attract customers and ensure that they return to our banks in the future



The accessibility of this product (it can be implemented into banking websites and apps) will entice customers who would normally reject the idea of meeting within a bank into thinking about applying for loans



This model speeds up decision turn over for a banking company and they will be less likely to accept applicants that are likely to default on payments in the future

Future Scope

1. Use model insights to improve accuracy and efficiency in loan decisions.
2. Refine the model by removing low-importance features to boost performance.
3. Collect additional variables like expenses, outstanding loans, and profession for better predictions.
4. Regularly update the model to maintain accuracy and adapt to new trends.

